MOSAM BISWAS

REVISION 3 · 6 NOVEMBER 2026

The complete technical anatomy of **www.mosambiswas.com**, a dark editorial portfolio built with zero frameworks and zero build step. Version six set the whole site on fire: a hand written WebGL ember field, an inertial motion system, three page specific choreographies. The first revision made it an *offline first* application. The second opened a door for the *terminal*: the bare domain answers **curl** and **ssh**, and the site hides seventeen easter eggs that keep their own score. This one traces every number on the work cards back to its repository.

(A) · STACK

HTML · CSS · Vanilla JS · GLSL · Go

(B) · HOSTS

GitHub Pages on www, a Go server on the bare domain

(C) · DEPENDENCIES

None on the site. Hand rolled.

(D) · EDITION

Vol. 06 · Rev. 3 · Nov 2026

(01) · OVERVIEW

ONE REPO, NO *machinery*

The site is a single static repository, `BiswasMosam.github.io`, served straight from GitHub Pages at **www.mosambiswas.com**. There is no bundler, no transpiler, no `package.json`. Every line that ships is a line that was written by hand: seven HTML pages, four stylesheets and ten small scripts.

What Vol. 06 brought. Three motion engines joined the v5 base: `shader.js` (a full page WebGL ember field), `motion.js` (inertial scroll, velocity shear, letter cascades), and two page specific systems, `cert-motion.js` for the certificates dossier and `sheichobi/darkroom.js` for the photography darkroom.

What the first revision added. An installable application that precaches itself in full, a résumé page in the same design system, and a contact form that refuses to lie about whether your message left. Sections 08 and 09.

What this revision adds. A second host. The bare domain is now a small Go server, Shellfolio, that answers `curl` with the résumé and `ssh` with the whole portfolio as an app (section 12). And a game: seventeen easter eggs, one tracker, one spoiler page (section 13).

> COUNTED, NOT ESTIMATED
>
> `3,247` lines of JavaScript that ship: ten files plus the inline scripts of the 404 page and `/help`. `0` dependencies, `0` build steps. The site is what you see in the repo. The server is counted separately, in section 12.

FILE	ROLE
`main.js`	Preloader gate, cursor and bubbles, reveals, parallax, clock, menu, contact form, modal, most eggs
`motion.js`	Inertial scroll, shear, cascades, marquee, magnetic buttons
`shader.js`	WebGL ember field, GLSL fragment shader
`cert-motion.js`	Dossier motion for `certificates.html`
`darkroom.js`	Darkroom motion for the SheiChobi gallery
`sheichobi.js`	Gallery: filters, load more, lightbox, collections, safelight
`service-worker.js`	Offline first cache. Precaches the whole shell at install
`sw-register.js`	Registration, update flow, dead origin eviction
`eggs.js` NEW	The one list of easter eggs, the score, the right click menu
`cv/cv.js` NEW	Turns the terminal résumé back into a page in a browser
`style.css`	Design system and every page level style
`fonts.css`	Self hosted `@font-face`, latin subsets
`humans.txt` NEW	Credits, in the humanstxt.org convention

FOUR PAGES IN THE MENU

`index.html` (portfolio) · `resume.html` · `certificates.html` · `sheichobi/sheichobi.html` (photography). All four are precached, so all four work offline.

THREE THAT ARE NOT NEW

`404.html`, which has a terminal in it · `/help`, every easter egg spoiled · `/cv`, the résumé for `curl`. None is in the sitemap.

"No frameworks, no build step. Every line that ships was written by hand."

(02) · DESIGN SYSTEM

WARM BLACK, BONE INK, *vermilion*

The palette is deliberately narrow: a warm near black canvas, an off white "bone" ink, and exactly one accent. Hierarchy is carried by two dimmed alpha steps of the ink rather than extra hues. Hairlines at 14% ink alpha do all the structural work, the same rule that frames this page.

The editorial voice comes from numbered section labels, `(01) · SELECTED WORK`, set in Space Mono, and from emphasised words dropped into *Instrument Serif italic* in accent colour, mid sentence, exactly as on this page.

HOUSE RULES

Easing: `cubic-bezier(0.22, 1, 0.36, 1)`, a long decelerating tail used by nearly every transition. Fluid padding: `clamp(1.25rem, 4vw, 4rem)`.

> **COPY RULE**
>
> No em or en dashes anywhere in visible copy. Ranges read `2022 / 2026`, separators are middots, and inline dashes became commas, colons or full stops. Ordinary hyphens in compounds stay. This document follows the same rule.

TOKEN	VALUE	USE
`--bg`	`#0b0b0a`	Warm black canvas
`--bg-2`	`#131311`	Raised surfaces
`--ink`	`#edebe4`	Primary text
`--ink-dim`	ink 58%	Body copy
`--ink-faint`	ink 34%	Meta text
`--line`	ink 14%	Hairline rules
`--accent`	`#ff5227`	Vermilion, the only colour

TYPE, SELF HOSTED

The four faces are served from `/fonts/` as latin subset woff2 files declared in `fonts.css`, because an offline application that falls back to Arial is not the same site. From this revision the document is set in them too: Syne for these headings, Manrope for this sentence.

FACE	ROLE
Syne 600/700/800	Display, wordmark, headings
Manrope 400/500/600	Body copy
Space Mono 400	Labels, meta, numerals
Instrument Serif 400*it*	Accent words, mid sentence

ONE LANGUAGE, FOUR DIALECTS

The tokens above are shared by every page without exception. What differs is the **motion vocabulary**: each page borrows a metaphor from the thing it is showing you. The rest of this document is organised around that split.

PAGE	DIALECT	GATE
index.html	Embers	has-motion
certificates.html	Dossier	has-dossier
sheichobi.html	Darkroom	has-darkroom
resume.html	Print	@media print

> **WHY GATES AND NOT SCRIPTS**
>
> Each dialect is a class its script puts on `<html>`, and every rule for it is written behind that class. A page with JavaScript disabled, or a visitor who asked for reduced motion, gets a stylesheet where those blocks simply never match. Nothing has to be undone, because *nothing was ever applied.*

(03) · THE HERO

EDGE TO EDGE, TO THE *pixel*

The hero name **MOSAM / BISWAS** is fitted to the exact container width by JavaScript rather than by a fixed type scale: each line is set at a 100px probe size, measured, then scaled by `containerWidth / wordWidth`. The second line is outlined via `-webkit-text-stroke` with a transparent fill.

Why one pass is not enough. Pixel based letter spacing and stroke widths do not scale with font size, so a single corrective pass always lands slightly wide. `fitLines()` runs up to three passes and breaks early once the measured width is within 1px of the true edge.

Because `motion.js` splits the hero into per letter spans after the first fit, it exposes the fitter as `window.__refitHero` and calls it again once the split is done. Resize is debounced onto the same function.

```
// main.js · fit each display line to its box
let size = 100;
for (let pass = 0; pass < 3; pass++) {
  const width = word.getBoundingClientRect().width;
  if (width <= 0) break;
  size *= line.clientWidth / width;
  line.style.fontSize = `${size.toFixed(2)}px`;
  if (Math.abs(width - line.clientWidth) < 1) break;
}

// motion.js re fits after the letter split
window.__refitHero = fitLines;
```

DETAIL

The same measure and scale trick fits the `LET'S TALK` contact title and the SheiChobi and certificates hero lines. One function, four pages.

THE SECTION HEAD GRAMMAR

Every section on every page opens with the same three part header, which is what makes four differently animated pages read as one document.

`__label`	Space Mono, accent, `(01) · Selected Work`
`__title`	Display, with one word in serif italic
`__aside`	Mono meta, a count or a range

Ranges read `2022 / 2026`, never with a dash. On narrow screens `.sec-head--stack` drops the aside under the title instead of letting it collide.

WHAT THE GRID DROPS

The work list is a CSS grid of five columns: index, title, category, meta and arrow. Rather than reflowing into cards, it **sheds columns** as the viewport narrows, in order of how much each one earns its space.

BELOW	DROPS
`1100px`	Category line
`900px`	Tech stack meta
`Always`	Index, title and arrow survive every width

That ordering has a consequence worth stating: **the title is the only column present at every width**, so any per project marker has to live there. It is why the `Live` badge is a `<sup>` on the title and not a column of its own.

MASKS, NOT OPACITY

Display lines sit inside `overflow: hidden` wrappers, so a heading can rise into place from below the baseline rather than fading in. The mask is part of the markup from the start, which means the same structure serves the v5 reveal, the v6 letter cascade and the dossier redaction sweep without any of them knowing about the others.

MEASURED, NOT TYPED

A fixed `clamp()` type scale would set the hero to a size that is nearly right at every width. Measuring sets it to the size that is *exactly* right at this one. The cost is three `getBoundingClientRect` calls on load and on resize.

(04) · THE EMBER FIELD

HAND WRITTEN GLSL, NO *libraries*

Behind every page sits a fixed `<canvas>` at `z-index: -1` running a raw WebGL fragment shader. No Three.js, no helper library, just a compiled program and a fullscreen triangle, three vertices, cheaper than a quad. The body background went transparent so the field shows through everything.

The image is **domain warped smoke**: a hash based value noise is stacked into four octaves of fBm, then fed into itself twice, `fbm(p + 2.1 · r)` where `r` is built from `q`, which is built from `p`. Bone coloured smoke rises from the folds; a smoothstep ridge extractor paints *vermilion embers* along the crests; a vignette pulls the edges back to editorial black; a one bit dither kills gradient banding.

THE ENERGY SYSTEM

The field breathes with the page. A single `uEnergy` uniform blends four inputs: hero presence (strong at the top), contact flare (strong at LET'S TALK), smoothed scroll velocity, and a `uBurst` ignition that fires at 1.4 when the preloader lifts and decays by ×0.982 per frame. Mid read, the field goes quiet.

CURSOR WARMTH

The mouse position is lerped (0.07) into a `uMouse` uniform; a Gaussian falloff `exp(-d²·7)` adds local heat, scaled by smoothed cursor speed. The warp itself is bent toward the pointer, so the smoke *leans into your hand.*

```
// domain warped fbm · the smoke
vec2 q = vec2(fbm(p*1.7 + t),
              fbm(p*1.7 - t*0.8 + 4.7));
vec2 r = vec2(fbm(p*1.7 + 2.3*q + ... + t*0.55),
              fbm(p*1.7 + 2.3*q + ... - t*0.4));

// the field leans toward the cursor
r -= (p - m) * exp(-md * 2.6) * 0.4;
float f = fbm(p*1.7 + 2.1*r);

// bone smoke + vermilion embers
col += vec3(.93,.92,.89) * f*f*f * 0.13
       * (0.35 + 0.65 * energy);
float ridge = smoothstep(0.50 - 0.24*uOver,
       0.86 - 0.22*uOver, f * (0.55 + 0.45*q.x));
col += vec3(1.0, 0.32, 0.15) * ridge
       * 0.42 * energy;
```

PERFORMANCE DISCIPLINE

- **0.5× render scale**, capped at 1.5 dpr. The field is soft by nature, so nobody can tell.
- **Context flags** `alpha:false`, `antialias:false`, `depth:false`, `powerPreference:'low-power'`.
- **Adaptive governor** counts frames over 27ms; after 45 slow frames it degrades resolution once more by ×0.65, permanently.
- **Never a dependency.** Under `prefers-reduced-motion`, or if WebGL or compilation fails, the canvas removes itself and the site falls back to plain `--bg`.

SEVEN UNIFORMS, AND WHAT THE PAGE PUTS IN THEM

The entire conversation between the document and the shader happens through seven numbers, uploaded once per frame. There is no other channel. The seventh arrived with the Konami code (section 13).

UNIFORM	CARRIES
uRes	Canvas size in device pixels
uTime REV	Accumulated seconds, not read off the clock, so a speed up never jumps. Scaled to 0.055 in the shader
uMouse	Lerped pointer position, y flipped to match GL
uMspd	Smoothed cursor speed, 0 to 1.2

UNIFORM	CARRIES
uEnergy	`0.2 + 0.85 · max(hero, tail) + velocity`, eased at 0.04
uBurst	Set to 1.4 once, on `body.is-loaded`, then ×0.982 per frame
uOver NEW	Overdrive, 0 to 1. Lowers the ember threshold for 3.2s

(05) · THE MOTION SYSTEM

THE PAGE LEANS INTO ITS OWN *momentum*

INERTIAL SCROLL

On fine pointer devices, wheel events are intercepted (`preventDefault`) and fed into a target value; a rAF loop eases the real scroll position toward it at lerp 0.11. Native jumps, keyboard and scrollbar drags, are detected when the actual position drifts more than 1.5px from the last write, and the system resyncs instead of fighting. Anchor links glide on the same easing. `html.smooth` disables native smooth scroll so the two never conflict.

VELOCITY SHEAR

Scroll velocity is smoothed (lerp 0.12) and mapped to a `skewY` of at most ±3.4° applied to the hero, section heads, work list, research and photography blocks. The page visibly shears with momentum and *settles when you stop.*

LETTER AND WORD CASCADES

The hero and contact titles are split into per letter `.hero__ltr` spans that rise from `translateY(118%) rotate(5deg)` inside the existing overflow masks, staggered 50ms per letter via a `--i` custom property. Section titles split into words (70ms stagger); work rows and photo strips enter with 60/90ms staggers under IntersectionObserver.

VELOCITY DRIVEN MARQUEE

The CSS keyframe marquee hands over to JS (`.is-js`): base drift 42px/s, plus a velocity term clamped at ±860. It accelerates as you scroll and reverses when you backtrack, wrapping by modulo on the half width of the duplicated track.

THE REST OF THE CHOREOGRAPHY

- **Header** hides past 280px when velocity › 5, returns on any upward motion.
- **Progress bar**, a 2px vermilion hairline scaled `scaleX(y / maxScroll)`.
- **Hero lines drift apart**: MOSAM slides left, BISWAS right, up to 220px as you leave the hero.
- **Magnetic buttons**: CTAs and the menu button lean toward the cursor (0.26x / 0.34y of the offset).
- **Stat counters** count up over 1.2s with cubic ease out, zero padded.
- **Scroll hint** fades out after 80px.

ONE GATE

Everything sits behind `html.has-motion`, which `motion.js` adds only when `prefers-reduced-motion` is off. With JS disabled or motion reduced, the stylesheet's v6 block is inert and the v5 reveals still work.

"Hand rolled, no libraries." *The header comment of every v6 file.*

(06) · BUBBLES

A CURSOR THAT WHISPERS & *reveals*

The signature interaction. One fixed `#cursor` div, a 12px white dot in `mix-blend-mode: difference`, lerp follows the pointer (factor 0.22) and morphs into three distinct instruments depending on what it touches. Desktop only (`pointer: fine`), removed entirely under reduced motion.

WHISPER MODE, WIRED TO SEPARATE WORDS

Any element carrying `data-whisper` morphs the dot into an ink coloured caption pill with Space Mono text. Crucially, whispers are wired to **individual words**: in "Developer, researcher, photographer" each word owns its own whisper, so sweeping the sentence produces a running commentary. A `||` delimiter stores alternate lines that cycle on every re hover, tracked via `data-whisperIdx`.

The pill drifts 26px above the pointer, flips below near the viewport top (`targetY < 60` gives `+34`), clamps at the horizontal edges, and sizes itself to label width + 36px by 32px. Links grow the dot to 52px instead, unless they whisper too.

REVISED · SEPTEMBER 2026

Whispers moved off the headings themselves and onto **inner spans**. A `<h2>` is a block box, so hovering the empty half of a headline used to fire the pill over blank space. Wrapping the words means the bubble only speaks where there is type to speak over.

LENS MODE, THE READING LENS

Paragraphs marked `.secret` carry two layers: the visible `.secret__base` and an absolutely positioned `.secret__layer`, alternate text set in accent coloured *Instrument Serif italic*. Over a secret, the dot expands into a 78px radius transparent circle with a vermilion border, and the hidden layer is unmasked through a per frame `clip-path: circle(r at x y)` that tracks the cursor. Move the lens, read the subtext.

```
layer.style.clipPath =
  `circle(${r}px at ${x}px ${y}px)`;
// lens radius lerp 0.16 · pill offset 0.2
```

WHERE IT SPEAKS

- **The version label**: "the smoke is hand written GLSL" and "no libraries, just math and embers". The site annotating its own shader.
- **The copyright line**, four rotating variants, including "inspect element is not a license".
- **The photo strip**, one word per frame: street, portrait, wildlife, monochrome, concert.
- **On SheiChobi** the bubble takes `.is-photo` (opacity 0) over photographs, so the difference blend never inverts an image.
- **And nowhere, for a while.** Thirty seconds without input and it falls asleep, snoring two small z's. One of the eggs in section 13.

The same lerped cursor position that drives the pill also feeds the shader's `uMouse`, so the smoke and the bubble move as *one hand.*

(07) · PRELOADER & LIFECYCLE

HELD AT 99 UNTIL IT'S *true*

The v5 preloader counted 00 to 100 on a fixed one second cubic ease and lifted its curtain regardless of what had actually loaded. In v6 the count is a real load gate: it holds at 99 until `document.fonts.ready` and the window `load` event have both resolved, so the curtain never lifts onto a page that still shifts. A 3.5s `MAX_WAIT` cap races the gate so slow networks cannot trap anyone.

```
Promise.race([
  Promise.all([fontsReady, pageLoaded]),
  cap  // setTimeout(resolve, 3500)
]).then(() => { assetsReady = true; });

if (!assetsReady) count = Math.min(count, 99);
```

When the curtain lifts (`translateY(-100%)`), `body.is-loaded` triggers the hero letter cascade, and the shader's ignition burst fires at the same moment. **Load, ignite, cascade: one choreographed instant.**

KEPT FROM V5

- **Film grain**, a fixed SVG `feTurbulence` data URI at 0.045 opacity over everything.
- **Scroll parallax**: `data-parallax` images translate and scale (1.12) against scroll.
- **Work list hover**: rows invert via a `scaleY` `::before` wipe, with a cursor following preview card at lerp 0.12.
- **Live IST clock**, `Intl.DateTimeFormat` in Asia/Kolkata, ticking every second.
- **Fullscreen mobile menu** with staggered links, and ASCII art in the source. The old right click guard and console line grew into eggs (section 13).

> **A NOTE ON THE COUNT**
>
> The gate is honest in both directions. If everything is already cached, which is the normal case once the app is installed, `assetsReady` resolves immediately and the count runs its full one second ease anyway. The curtain is a *promise about layout*, not a progress bar.

THE ORDER THINGS HAPPEN

Five scripts, no module system, no bundler, and therefore no dependency graph to resolve. Order is expressed the old way: by where the `<script>` tags sit, all of them at the end of `<body>`. The one flag that has to run early, `html.js`, is set inline in the `<head>` so the no JavaScript stylesheet never flashes.

#	MOMENT	WHAT HAPPENS
01	`<head>`	`html.js` is set inline; `fonts.css` and `style.css` are requested
02	`eggs.js`	Defines `window.eggs` first, so every later script can report a find
03	`main.js`	Preloader count starts; `fitLines()` measures the hero
04	`motion.js`	Adds `html.has-motion`, splits the hero into letters, calls `__refitHero`

#	MOMENT	WHAT HAPPENS
05	`shader.js`	Compiles the program, mounts the canvas, or removes it and gives up
06	`fonts + load`	Both resolve, or 3.5s passes. `assetsReady` flips
07	`count hits 100`	150ms, then `body.is-loaded`: the cascade runs and `uBurst` fires at 1.4, together
08	`sw-register.js`	On `load`: registers, or tears down a worker on a dead origin

(08) · THE OFFLINE APPLICATION

A ROOM WITH NO *internet*

The brief for this one was literal: in a room with no internet, the whole portfolio, the résumé, the certificates and **the type it is set in** have to already be on the phone. That is a stricter requirement than "has a service worker", and it changed four things.

PRECACHED IN FULL, ATOMICALLY

The earlier worker filled its cache in as pages happened to be visited. This one declares a `SHELL` of everything needed to render all four pages with the network switched off, and installs it with a single `cache.addAll()`. That call is all or nothing on purpose: **if one entry is wrong the install fails loudly**, rather than leaving a half usable app that only breaks once you are already offline.

TIER	CONTENTS
`SHELL`	45 entries: 4 pages, 3 stylesheets, 8 scripts, manifest, icons, 3 PDFs, 16 font files. Atomic.
`MEDIA`	Certificates, their six résumé thumbnails, feature photographs. Added one at a time, allowed to fail.
`Thumbs`	Read from `photos.json` at install, so adding a photograph never means editing the worker.

Full size photographs are deliberately left out. They are roughly 40 MB, and they are cached individually as they are opened instead.

TYPE IS PART OF THE SHELL

All sixteen woff2 subsets ship in `SHELL`. This is why the fonts were brought in house in the first place: an offline site that silently falls back to Arial is not *this* site.

DEAD ORIGIN EVICTION

`sw-register.js` registers only on `mosambiswas.com` and on localhost, which is treated as production so the offline behaviour can be tested before it ships. **Anywhere else**, any previously registered worker is unregistered and every cache deleted. That eviction is not theoretical: it is what emptied the old `mosambiswas.me` app when the domain lapsed.

TWO STRATEGIES, CHOSEN PER DESTINATION

Network first for navigations and for scripts and styles, so the newest deploy wins whenever there is a network and the cache is only what is left when there is not. **Cache first** for fonts, images and PDFs, which cannot change without a redeploy, and a redeploy changes the cache name anyway.

```
if (request.method !== 'GET') return; // form POST
if (new URL(request.url).origin
    !== self.location.origin) return;  // third party

const isNavigation = request.mode === 'navigate'
  || request.destination === 'document';
const isCode = request.destination === 'script'
  || request.destination === 'style';

if (isNavigation || isCode)
  return event.respondWith(
    networkFirst(request, isNavigation));
event.respondWith(cacheFirst(request));
```

An offline deep link to a page that was never cached falls back to `/index.html`, so a visitor gets the portfolio rather than the browser's dinosaur. The contact form's POST and every cross origin request are left entirely alone.

Reload only to replace. REV The page used to reload on every `controllerchange`, including the very first `clients.claim()`, so every first time visitor got a surprise reload that cut the preloader. It now reloads only if the page already had a controller, which means a real update.

THE ONE DEPLOY RULE

`VERSION` at the top of the worker is bumped on **every** deploy. Currently `v36`, up from v18 at the first revision. The name change is the entire eviction mechanism; forget it and visitors keep the old site.

(09) · RÉSUMÉ, CONTACT, PROOF

BUILT FOR THE PERSON WHO HAS *four minutes*

A friend reviewed the site and the verdict was blunt: it was beautiful and it was slow to answer the only question a recruiter asks. This section is what came out of that review.

RESUME.HTML

A web résumé in the site's own design system, reusing `style.css`, `shader.js` and `main.js` with page specific styles inline. Seven sections, from where I've *worked* to beyond the *code*, a `@media print` stylesheet that strips to black on white, and two PDF downloads.

Certificates you can see. NEW The six certifications reuse the work list's cursor card, now one function, `bindPreview()`, serving both lists. Hovering a row shows the certificate as a WebP thumbnail, 720px on its long side; the card opens left of the cursor when the right edge is too close. Clicking opens the full certificate in the shared modal.

One résumé, five renderings. The one page black and white PDF, typeset in LaTeX, is the source of truth. The colour PDF, this page, the terminal résumé and `/cv` are all kept to it.

THE CONTACT FORM THAT WILL NOT LIE

`LET'S TALK` is a `role="button"` heading that morphs into a name, email and message form in place, closable with Escape. It posts by `fetch` to FormSubmit's AJAX endpoint with a honeypot field and no captcha.

The bug worth recording. FormSubmit answers `200` even when it refuses a message, and says so only in the body. Branching on `res.ok` alone showed visitors "Sent ✓" for mail that never left.

```
const body = await
res.json().catch(() => null);
if (!body || String(body.success)
!== 'true')
  throw new Error(body?.message
|| 'rejected');
```

Failure is not silent either: it points at the email address below, which is **click to copy** via the Clipboard API, falling back to `mailto:` when the API is missing or throws.

WORK ROWS CARRY OUTCOMES

Hovering a project no longer shows a name tile. It shows an **outcome stat card**: the index and year, one metric, and a single line of what it did. Every number can be checked in the project's repository; one with nothing behind it there does not go on a card.

Where a project is deployed, the row links to the *running app* rather than the repository, and the title carries a `Live` badge, or `Install` for something you install. The title is the only grid column that survives every breakpoint, which is why the badge lives there.

TWELVE ROWS, ONE NEW

`01 · AMINAL`, a local first voice assistant for Windows, leads the list. `02 · Locals` NEW, AI users that test your apps, follows it, and `03 · Fill.ai`'s row opens the download page.

THE TWELVE CARDS ONE NUMBER AND ONE LINE EACH, EVERY NUMBER TRACED TO ITS REPOSITORY

PROJECT	STAT	OUTCOME
AMINAL	Local	Voice, vision and memory on your GPU
Fill.ai	121	End to end checks, run on every push
Sim Sim	23	Voice commands in a PySide6 GUI
Rail Mitra	Zero	Conflict schedules, by constraint solving
MindFirst	7	Condition tests behind a GAD-7 pre-screen
PixelForge	0.89	ROC-AUC, leukemia detection

PROJECT	STAT	OUTCOME
Locals	114	AI users from nine job families
IntelliStatement	LLM	Extraction from PDFs and images
Committee Connect	Live	Real time Firebase sync
PixelShift	Zero	Uploads. It all runs in the browser
Flex Card	PNG	Share ready hardware spec cards
To-Do	E2E	Task boards with auth, end to end

(10) · CERTIFICATES: THE DOSSIER

PROOF OF WORK, *declassified*

The certificates page gets its own motion language, paper, stamps and filing cabinets, gated behind `html.has-dossier`. Three featured credentials, eight in the archive.

REDACTION BAR HERO

The title PROOF OF WORK arrives like a declassified document: solid bars, ink for the first line, vermilion for the outlined second, sweep across each line (`scaleX` 0 to 1 from the left, then 1 to 0 from the right), and the text snaps visible mid sweep, exactly when the bar covers it.

```
@keyframes dossier-sweep {
  0%   { transform: scaleX(0); origin: 0 50%; }
  46%  { transform: scaleX(1); origin: 0 50%; }
  54%  { transform: scaleX(1); origin: 100% 50%; }
  100% { transform: scaleX(0); origin: 100% 50%; }
}
```

STAMPED FEATURES

Featured certificates land with an overshoot bezier `(0.2, 0.9, 0.3, 1.15)` from `scale(1.05) rotate(0.6deg)`, stamped onto the page, followed by a decaying vermilion stamp flash `box-shadow`. The modal image reuses the same stamp in.

FILED TILES AND TYPED LABELS

Archive tiles file in one by one, 55ms stagger, each with a slight index derived rotation `(--i - 3) × 0.35deg`, like papers squared off by hand. Section labels type themselves character by character with a ▌ caret at 26ms per glyph; the "08 certificates" aside counts itself up. All triggered by IntersectionObservers.

PAPER TILT AND SHEEN

On desktop, every certificate tilts in 3D under the cursor, `perspective(720px)` with up to 9°/7° of rotation, while a radial sheen follows the pointer via `--gx` and `--gy` custom properties, like *light raking across a glossy print.*

```
el.style.setProperty('--ry',
  `${((px - 0.5) * 9).toFixed(2)}deg`);
el.style.setProperty('--rx',
  `${((0.5 - py) * 7).toFixed(2)}deg`);
```

WHAT IS IN THE CABINET

Three credentials are featured with their own panel and a stamp in animation. The remaining eight are archive tiles, filed in sequence.

THE BIG THREE

01 Research presentation, ICAIIHI 2025
02 Software development internship, Sedna Technologies
03 Samsung Innovation Campus, AI

EVERYTHING ELSE

01 Android Compose Camp
02 AWS Machine Learning
03 Generative AI
04 Agile and Scrum fundamentals
05 Industry focused Java workshop
06 Programmable circuit building
07 Hacktoberfest
08 Smart India Hackathon 2025

(11) · SHEICHOBI: THE DARKROOM

PRINTS THAT DEVELOP AS YOU *watch*

Shei Chobi, Bengali for "that picture", is the photography wing: 84 photographs in six collections, and in v6 it behaves like a darkroom, light, chemistry and optics, gated behind `html.has-darkroom`.

THE DEVELOPING BATH

The hero title **SHEI CHOBI** arrives overexposed, `blur(18px) brightness(2.3)`, and settles into focus over 1.7s, like a print in developer fluid. Every archive photo goes through the same bath: from `blur(14px) brightness(1.9) grayscale(1)` to full colour, batch staggered 90ms. Because the grid is rebuilt on every filter and load more, a MutationObserver re watches new cards automatically.

APERTURE LIGHTBOX

Clicking a photo opens the lightbox as an aperture: a `clip-path: circle()` expands from 0% to 141% centred on the exact click point, captured into `--cx` and `--cy`. Between photos, a shutter blink, brightness flash and blur, restarted by forcing reflow when the image `src` mutates.

FOCUS PULL

Hovering the contact sheet racks focus like a lens: the hovered frame stays at full strength while every sibling falls back to `opacity: 0.4`. A pure CSS `:hover` chain, desktop only. It used to blur every sibling, which repainted the whole sheet on each hover.

> REVISED · WHAT THE LAG TAUGHT
>
> The grain originally breathed by applying a **filter to the sheet itself**, up to 2.6px of motion blur. That forces the browser to re rasterize the entire grid every frame, and scrolling janked on exactly the devices that matter. The loop now animates **opacity only**, and it parks itself the moment velocity settles instead of running forever.

THUMBNAILS, NOT ORIGINALS

The second half of the fix was weight. The archive grid and the collection previews load 900px wide webp thumbnails from `Photographs/thumbs/`; only the lightbox loads the original. A small Node script regenerates them, and a photograph without a thumbnail still works, it just loads slower until the script is run.

```
// generate-thumbs.js · run after adding photos
await sharp(src)
  .resize({ width: 900, withoutEnlargement: true })
  .webp({ quality: 82 })
  .toFile(dest);
```

The same `photos.json` that drives the grid is what the service worker reads at install time to decide which thumbnails to precache.

"Every page speaks the same design language, but each one moves in its own dialect: embers, dossiers, darkrooms."

(12) · THE SECOND DOOR

THE PORTFOLIO, IN A *terminal*

Two commands work on the bare domain: `curl mosambiswas.com` prints the résumé in colour, and `ssh mosambiswas.com` opens the whole portfolio as an app. GitHub Pages can do neither, since it only serves files, so on **27 September 2026** the bare domain moved to a server of its own, **Shellfolio**, one Go binary on a small VPS. The website did not move at all.

TWO RECORDS, TWO HOSTS

```
mosambiswas.com       A      → VPS, Shellfolio
  :22    Wish SSH     → the Bubble Tea app
  :80    curl, wget   → the résumé as text
  :443   browsers     → 301 to www, path intact
www.mosambiswas.com   CNAME → GitHub Pages
```

The apex used to point at GitHub Pages only to redirect to www. Shellfolio kept that redirect identical, path and query included, so old links and the installed app on www never noticed. HTTPS comes from Go's `autocert`; port 80 stays plain HTTP, because that is what a bare `curl` speaks.

WHO COUNTS AS A TERMINAL

The handler reads the User-Agent. Clients that print a body straight into a terminal get text; everyone else gets the redirect. Every response says `Vary: User-Agent`, so no cache in between can hand a browser the text.

```
terminalClient = regexp.MustCompile(`(?i)^(curl|
  wget|httpie|xh|fetch|libfetch|aria2|lwp-request)\b`)
// not a terminal: send it on to www
http.Redirect(w, r, www+r.URL.RequestURI(), 301)
```

PATH	RETURNS
`/` · `/cv`	The résumé in 256 colour, 78 columns wide
`/plain`	The same with no escapes, for saving to a file
`anything else`	A short text 404 that points at the two commands

THE APP OVER SSH

Wish runs the SSH server; Bubble Tea draws a full screen app in four tabs: **Work**, all eleven projects, **About**, **Résumé** and **Contact**. It is keyboard only, so visitors can still select text, and `c` copies a link or the email to their own clipboard through OSC 52. A session with no terminal gets the plain résumé.

BUILT FOR STRANGERS

- **No shell, ever.** Every session runs the app or prints text.
- **No login.** No password or key asked for, none stored.
- **Rate limited** per IP, one new session every 2s with bursts of 5; at most 64 open at once, closed after 15 idle minutes and after an hour regardless.
- **Unprivileged.** The systemd unit grants one capability, `CAP_NET_BIND_SERVICE`, and the filesystem is read only to it apart from `/var/lib/shellfolio`.

THE HONEST EXCEPTION

The site still has zero dependencies. The server does not: Wish, Bubble Tea and Lip Gloss are Charm's libraries, and a hand written SSH server would be bravado, not craft. `1,467` lines of Go, `155` of them tests: every line of every tab must fit 44 to 140 columns, and no em or en dash may appear in the copy.

/CV · THE SAME TRICK WITH NO SERVER

Before the server, `curl -L www.mosambiswas.com/cv` ran on GitHub Pages alone, and still does. `cv/index.html` is one file that is two things: its HTML head hides inside an OSC escape, `ESC ] 9999 ; … BEL`, that terminals swallow unprinted, so curl shows only the coloured résumé; a browser reads the same bytes as HTML, and `cv.js` turns the colour codes into styled spans. Shellfolio generates it from the same `content.go` the server reads.

(13) · EASTER EGGS

SEVENTEEN THINGS NOBODY *asked for*

The site already had a few jokes in it: a console line, a portrait in the source, the whispers. On 26 and 27 September they became a **collection**: seventeen eggs across the site, its project pages and two terminals, a tracker that remembers which ones this browser has found, and a spoiler page, `/help`, for anyone who gives up.

ONE LIST, ONE SCORE

`eggs.js` loads before `main.js` on every page and holds the only list. Finding an egg writes a timestamp to `localStorage` under `mb:eggs`, and the first time, a small note slides in, `Easter egg found · 05 / 17`, with a link to see how many are left. Nothing is sent anywhere. The note's styles are injected by the script, so it looks the same on pages whose CSS it has never met.

```
window.eggs = {
  list,          // the seventeen, in /help's order
  found,         // { id: when }, this browser only
  find,          // record once, announce once
  reset,         // /help's two click "forget"
  contextMenu    // the right click menu
};
```

CLAIMS, WHERE NO SCRIPT RUNS

Four eggs live where JavaScript cannot see them: the page source, `humans.txt`, and the two terminals. Each ends with a link home, `/?source`, `/?humans`, `/?curl`, `/?ssh`, and arriving through it counts. The homepage records the claim at once, cleans the URL, and parks the note in `sessionStorage` until the preloader has lifted, so even a reload in between still says found.

A RIGHT CLICK MENU OF ITS OWN

Right click used to be switched off. It now opens a small menu: the source, the credits, copy my email, and the egg count. It is a real `role="menu"` with arrow keys and Escape, and text fields keep the browser's own menu, paste and spelling included.

SHARED ACROSS THE PROJECTS

PixelShift and Fill.ai are served from the same origin, under `/PixelShift/` and `/Fill.ai/`, so they share this browser's storage. Both load `/eggs.js` and report their own egg into the same count.

RULES THAT KEEP IT FAIR

- **Only a real key press earns the Konami egg.** `/help` has a "type it for me" button that dispatches the same key events. `e.isTrusted` is false for those, so the flare runs and the egg does not count.
- **A wrong case is not being lost.** GitHub Pages paths are case sensitive and it has no redirect settings, so the 404 page forwards `/fill.ai/` to `/Fill.ai/` before it paints, and sets `window.__forwarding` so the lost egg is not credited on the way through.
- **First visits keep their claim.** The old first visit reload (section 08) used to swallow `/?curl` for exactly the people most likely to arrive through it.
- **Spoilers are opt in.** `/help` is `noindex`, out of the sitemap, and linked only from places a curious person ends up: the 404 terminal, `humans.txt`, the right click menu, the console.

THE /HELP PAGE

A shelf of seventeen squares that fill in as eggs are found, a switch that blurs the steps for the ones not found yet, a two click reset, and a live try it for most of them: Konami keycaps that light up as you type, a whisper and a lens to hover, the source portrait fetched on demand.

"You read the credits, so this file is an egg as well." *The last lines of humans.txt.*

(13) · EASTER EGGS, CONTINUED

WHERE THEY *hide*

#	EGG	WHERE	TRIGGER
01	Konami code	anywhere	↑↑↓↓←→←→BA
02	The whispers	headings	hover
03	The lens	.secret text	hover
04	The bubble sleeps	anywhere	30s still
05	Type a word	homepage	hire · mosam
06	Right click	anywhere	right click
07	Console note	dev tools	egg()
08	Print the homepage	homepage	Ctrl P
09	Source portrait	index.html	view source
10	humans.txt	/humans.txt	read it
11	The 404 terminal	any bad path	get lost
12	Bots get answers	/wp-admin, /.env	type one
13	Darkroom safelight	SheiChobi	R
14	PixelShift, split	/PixelShift/	rgb
15	Fill.ai, once more	/Fill.ai/	Alt Shift F
16	The résumé by curl	a terminal	curl
17	The portfolio by ssh	a terminal	ssh

THE 404 TERMINAL

GitHub Pages serves `404.html` for a missing path at any depth, so every URL on it is absolute. It sets the hero's solid over outline type on the ember field, **404 / LOST**, and adds a small working shell: `help`, `ls`, `cd`, `cat humans.txt`, `whoami`, tab completion and ↑↓ history.

It reads the path that was asked for and, by Levenshtein distance, guesses the page that was meant: `/resme` gets *did you mean /resume.html?* Every line is built with `textContent`, never `innerHTML`, because the path and the commands come from whoever is typing.

Seven patterns answer the paths bots probe all day: WordPress, `.env`, `.git`, admin panels, `*.php`, APIs, keys and backups. Bots read only the status code. The answers are for the one human who tries by hand.

KONAMI: FORGE OVERDRIVE

`main.js` dispatches `forge:overdrive` and `shader.js` raises `uOver` for 3.2 seconds. Raising the ignition burst alone was nearly invisible, because embers come from a smoothstep *threshold,* not from brightness. So overdrive moves the threshold, and far more of the smoke catches.

```
smoothstep(0.50 - 0.24*uOver,
  0.86 - 0.22*uOver, ...)
clock += dt * (1.0 + 4.0*over);
```

The smoke churns five times faster, and because time is accumulated rather than read off the clock, the speed up never makes it jump. With reduced motion the code is still acknowledged, and the embers stay calm, *as asked.*

THREE TRAPS, RECORDED

- **The sleeping bubble breathes by size, never by scale.** JS writes the bubble's `transform` every frame, and a CSS `scale` multiplies that translate too. It dragged the bubble about 140px toward the corner.
- **The safelight is a layer, not a filter.** One fixed layer of `#ff2200` in `multiply`: white turns red, black stays black. Section 11 already showed what a filter over the photographs would cost.
- **The print card is pure CSS.** Only the homepage carries a `.print-card`, and `body:has(> .print-card)` hides everything else on paper. Its QR code is inline SVG, generated once. No library runs.

"There is no admin panel. The admin is me, and I use VS Code." *The 404 page, to anyone who tries /admin.*

(14) · ENGINEERING LEDGER

FAST, ACCESSIBLE, *honest*

PERFORMANCE DECISIONS

- **Zero dependencies on the site.** No framework tax; 3,247 lines of vanilla JS ship. The Go server, which never runs in a browser, is the one place a library earns its keep.
- **Transform and opacity only.** Every animation runs on compositor friendly properties; staggers via `--i` custom properties, not per element JS timers.
- **One rAF loop per system.** Motion, shader and darkroom each run a single frame loop; observers do the triggering, and the darkroom loop parks itself when idle.
- **Adaptive shader quality**, 0.5× render scale, low power context, self degrading on slow frames.
- **Thumbnails in the grid**, originals only in the lightbox.
- **Preloader as load gate**, no layout shift after the curtain.
- **No blur over the canvas.** The 404 terminal sits on a flat fill, not `backdrop-filter`, which would re rasterize the box over a live WebGL canvas every frame.

ACCESSIBILITY & RESILIENCE

- **Reduced motion.** All four v6 engines bail on `prefers-reduced-motion`; the shader canvas removes itself; CSS gates make the v6 styles inert without JS.
- **No JS fallback.** Reveals are gated behind `html.js`; content is fully readable without a single script.
- **Skip link, `sr-only` h1, semantic landmarks, keyboard reachable modals and form.**
- **Honest status.** The contact form reports what actually happened, and points at a copyable address when it fails.
- **Menus that are menus.** The right click menu has a role, arrow keys and Escape, and never replaces the menu in a text field.

SEO & DISTRIBUTION

- **JSON-LD**, WebSite and Person structured data, OG and Twitter cards, sitemap and robots.txt.
- **Installable**, manifest with maskable icon, screenshots and three shortcuts: Résumé, Certificates, Photography.
- **Two hosts**: a CNAME for www on GitHub Pages, an A record for the bare domain on Shellfolio, which redirects browsers exactly as Pages did.
- **Credits** in `humans.txt`, linked from every page with `rel="author"`.

CONTENTS OF RECORD

11 Projects: AMINAL, Fill.ai, Sim Sim, Rail Mitra, MindFirst, PixelForge, IntelliStatement, Committee Connect, PixelShift, Flex Card, To-Do. Two live apps, one to install.

11 Certificates, three featured and eight archived.

84 Photographs in six collections.

01 IEEE publication: psychometric ML screening, ICAIIHI 2025. DOI 10.1109/ICAIIHI67124.2025.11403838

07 Pages. The four in the menu work offline.

17 Easter eggs, one score per browser.

02 Doors: a browser, and a terminal.

(•) · CONTACT

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(15) · RECOGNITION

OUT IN THE *wild*

On 10 July 2026 the site was submitted to **Dark Mode Design** (darkmodedesign.com), a curated directory of sites built dark by default, not dark mode as a toggle. The submission note doubled as a one paragraph brief of the entire project, reproduced here exactly as sent, em dashes and all:

> *"Warm-black editorial design — #0b0b0a with a single vermilion accent — hand-built with zero frameworks. Behind every page runs a hand-written GLSL ember field; the custom cursor whispers captions on individual words and turns into a lens that reveals the page's hidden subtext. Full dark from the first pixel — dark mode isn't a toggle here, it's the material."*

SUBMISSION

Directory	Dark Mode Design, darkmodedesign.com
Submitted	10 Jul 2026
Pitch covers	Zero framework build, hand written GLSL ember field, whisper and lens cursor

OUTCOME

Status	Accepted
Reviewer	Cai Cardenas
Queued live	Within about a week of acceptance, response received 20 Jul 2026

The pitch and the product turned out to be the same file: every claim in that one paragraph is documented, page by page, in the sections before this one. What the paragraph did not promise is what the two revisions added since: all of it *still works with the network switched off,* and it now answers in a terminal too.

DOCUMENT HISTORY

EDITION	COVERS
Vol. 06 · Jul 2026	Ember field, motion system, dossier and darkroom. Eleven sections.
Rev. 1 · 23 Sep 2026	Offline application, résumé and contact surface, thumbnail pipeline, whisper scoping. Thirteen sections.
Rev. 2 · 28 Sep 2026	Shellfolio and `/cv`, seventeen easter eggs, Fill.ai, certificate previews. Set in the site's own type. Fifteen sections.
Rev. 3 · 6 Nov 2026	Stat cards traced to their repositories: Fill.ai 121, PixelForge 0.89, MindFirst and Sim Sim by what they do. Locals added.

HOW TO VERIFY ANY CLAIM HERE

Every number in this document was read out of a repository rather than remembered. The shader constants are in `shader.js`, the cache tiers in `service-worker.js`, the eggs in `eggs.js`, the server's limits in the Shellfolio repository. Nothing is minified, so **view source is the primary source.**